In [1]:
```
# Exp :- 10
```

In [2]:
```
# Aim : To perform Operation Simple Linear Regression for Salary Data(SLR
```

In [10…
```
#Name:Arya patil
#Roll:42
#sec:C
#subject:DSS
#Date:
```

In [11…
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

In [12…
```
import os
```

In [13…
```
os.getcwd()
```

Out[13]:
```
'C:\\Users\\aryap'
```

In [14…
```
os.chdir("C:\\Users\\aryap\\Downloads")
```

In [15…
```
data=pd.read_csv("Salary.csv")
```

In [16…
```
data.head()
```

Out[16]:

| | YearsExperience | Salary |
|---|---|---|
| **0** | 1.1 | 39343 |
| **1** | 1.3 | 46205 |
| **2** | 1.5 | 37731 |
| **3** | 2.0 | 43525 |
| **4** | 2.2 | 39891 |

In [17…
```
data.tail()
```

Out[17]:

| | YearsExperience | Salary |
|---|---|---|
| **30** | 11.2 | 127345 |
| **31** | 11.5 | 126756 |
| **32** | 12.3 | 128765 |
| **33** | 12.9 | 135675 |
| **34** | 13.5 | 139465 |

In [18…
```
data.head(30)
```

Out[18]:

| | YearsExperience | Salary |
|---|---|---|
| **0** | 1.1 | 39343 |
| **1** | 1.3 | 46205 |
| **2** | 1.5 | 37731 |
| **3** | 2.0 | 43525 |
| **4** | 2.2 | 39891 |
| **5** | 2.9 | 56642 |
| **6** | 3.0 | 60150 |
| **7** | 3.2 | 54445 |
| **8** | 3.2 | 64445 |
| **9** | 3.7 | 57189 |
| **10** | 3.9 | 63218 |
| **11** | 4.0 | 55794 |
| **12** | 4.0 | 56957 |
| **13** | 4.1 | 57081 |
| **14** | 4.5 | 61111 |
| **15** | 4.9 | 67938 |
| **16** | 5.1 | 66029 |
| **17** | 5.3 | 83088 |
| **18** | 5.9 | 81363 |
| **19** | 6.0 | 93940 |
| **20** | 6.8 | 91738 |
| **21** | 7.1 | 98273 |
| **22** | 7.9 | 101302 |
| **23** | 8.2 | 113812 |
| **24** | 8.7 | 109431 |
| **25** | 9.0 | 105582 |
| **26** | 9.5 | 116969 |
| **27** | 9.6 | 112635 |
| **28** | 10.3 | 122391 |
| **29** | 10.5 | 121872 |

In [19…
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 35 entries, 0 to 34
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  35 non-null     float64
 1   Salary           35 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 688.0 bytes
```

In [20…
```
data.describe()
```

Out[20]:

| | YearsExperience | Salary |
|---|---|---|
| **count** | 35.000000 | 35.000000 |
| **mean** | 6.308571 | 83945.600000 |
| **std** | 3.618610 | 32162.673003 |
| **min** | 1.100000 | 37731.000000 |
| **25%** | 3.450000 | 57019.000000 |
| **50%** | 5.300000 | 81363.000000 |
| **75%** | 9.250000 | 113223.500000 |
| **max** | 13.500000 | 139465.000000 |

In [21…
```
data.shape
```

Out[21]:
```
(35, 2)
```

In [22…
```
data.size
```

Out[22]:
```
70
```

In [23…
```
data.ndim
```

Out[23]:
```
2
```

In [24…
```
data.isnull().sum()
```

Out[24]:
```
YearsExperience    0
Salary             0
dtype: int64
```

In [25…
```
#Independent and Dependent Variables
#x = data['YearsExperience']
#y = data['Salary']
```

# Independent and Dependent Variables

In [26…
```
X=data.drop('Salary',axis=1)
```

In [27…
```
X.head()
```

Out[27]:

| | YearsExperience |
|---|---|
| **0** | 1.1 |
| **1** | 1.3 |
| **2** | 1.5 |
| **3** | 2.0 |
| **4** | 2.2 |

In [28…
```
y=data.Salary
```

In [29…
```
y.head()
```

Out[29]:
```
0    39343
1    46205
2    37731
3    43525
4    39891
Name: Salary, dtype: int64
```

In [30…
```
from sklearn.model_selection import train_test_split
```

In [31…
```
X_train,X_test,y_train,y_test=train_test_split(X,y,random_state=0,test_si
```

In [32…
```
print(X_train.shape)
```
```
(24, 1)
```

In [33…
```
X_test.shape
```

Out[33]:
```
(11, 1)
```

# Model Fitting

In [34…
```
from sklearn.linear_model import LinearRegression
LR=LinearRegression()
LR.fit(X_train,y_train)
```

Out[34]:
```
LinearRegression()
```

In [35…
```
#Assigning Coefficient (slope) to m
m = LR.coef_
```

In [36…
```
print("Coefficient  :" , m)
```

```
Coefficient  : [8528.00205699]
```

In [37…
```
#Assigning Y-intercept to a
c = LR.intercept_
```

In [38…
```
print("Intercept : ", c)
```

```
Intercept :  30653.80404187284
```

# y=mx+c

# Prediction

In [39…
```
y_pred=LR.predict(X_test)
```

In [40…
```
y_pred
```

Out[40]:
```
array([120197.8256403 ,  88644.21802942,  74146.61453254, 118492.225228
9 ,
        98025.02029212,  72441.01412114,  63913.01206415,  43445.807127
36,
        64765.81226984, 112522.623789  , 107405.82255481])
```

In [41…
```
y_test
```

Out[41]:
```
29    121872
20     91738
16     66029
28    122391
22    101302
15     67938
10     63218
2      37731
11     55794
27    112635
25    105582
Name: Salary, dtype: int64
```

# Evaluation metrics

In [42…
```
from sklearn import metrics
```

Accuracy

In [43…
```
Accuracy = LR.score(X_test, y_test)
Accuracy
```

Out[43]: 0.9714499470616078

MAE

In [44…
```
print(metrics.mean_absolute_error(y_test,y_pred))
```

3807.469971826513

MSE

In [45…
```
print(np.sqrt(metrics.mean_squared_error(y_test,y_pred)))
```

4685.185783311836

RMSE

In [46…
```
rmse = np.sqrt(metrics.mean_squared_error(y_test,y_pred))
print("RMSE:", rmse)
```

RMSE: 4685.185783311836

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]: